

Discrete Math for Testers

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Propositional Logic

Symbol	Name	Description	Example
$\wedge$	AND	Both arguments must be true	$A \wedge B$
$\vee$	OR	One argument must be true	$A \vee B$
$\neg$	NOT	Opposite of argument	$\neg A$
$\rightarrow$	IMPLIES	Left indicates right is true	$A \rightarrow B$
$=$	EQUALS	True iff both are equal	$A = B$
$\oplus$	XOR	<i>Only</i> one must be true	$A \oplus B$

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Propositional Logic

A	B	$A \wedge B$	$A \vee B$	$\neg A$	$A \rightarrow B$	$A = B$	$A \oplus B$
False	False	False	False	True	True	True	False
False	True	False	True	True	True	False	True
True	False	False	True	False	False	False	True
True	True	True	True	False	True	True	False

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Propositional Logic



- What does $(\text{true} \wedge (4 < 2)) \vee a = a$ equal?
 - A. True
 - B. False
 - C. Undefined / Unknown

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Set Theory

- Set Definition:
 - Months = { January, February, March, April, ..., December }
 - Even = { $e: (e \bmod 2) = 0$ }
 - 613 Students = { student: student is enrolled in CIS 613 }
- Set Membership:
 - In a set: April $\in$ Months
 - Not in a set: Tuesday $\notin$ Months
- The Empty Set:
 - $\emptyset = \{\}, \{\emptyset\} = \{\{\}\}, \{\{\emptyset\}\} = \{\{\{\}\}\}$

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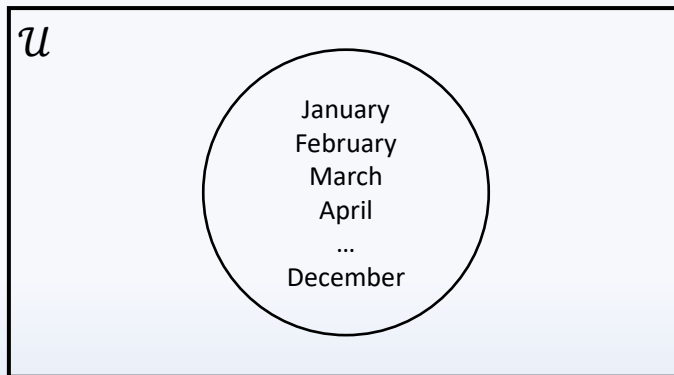
Propositional Logic



- If no students enroll in CIS 613, what is the following set equal to? Set is 613 Students = { student: student is enrolled in CIS 613 }
- A. $\{\{\}\}$
 - B. $\{\{\emptyset\}\}$
 - C. $\emptyset$
 - D. $\{\emptyset\}$

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Set Theory: Venn Diagrams



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Set Theory: Operations

- Typical Set Operations (On sets A and B):
 - Union: $A \cup B = \{x: x \in A \vee x \in B\}$
 - Intersection: $A \cap B = \{x: x \in A \wedge x \in B\}$
 - Complement: $A' = \{x: x \notin A\}$
 - Reflexive Complement: $A - B = \{x: x \in A \wedge x \notin B\}$
 - Symmetric Difference: $A \oplus B = \{x: x \in A \oplus x \in B\}$

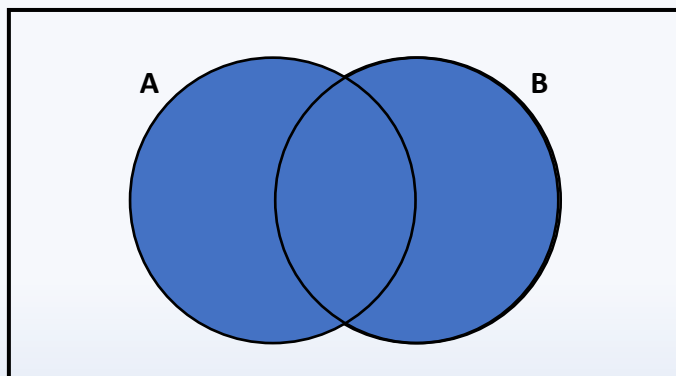
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Set Theory: Operations, Union



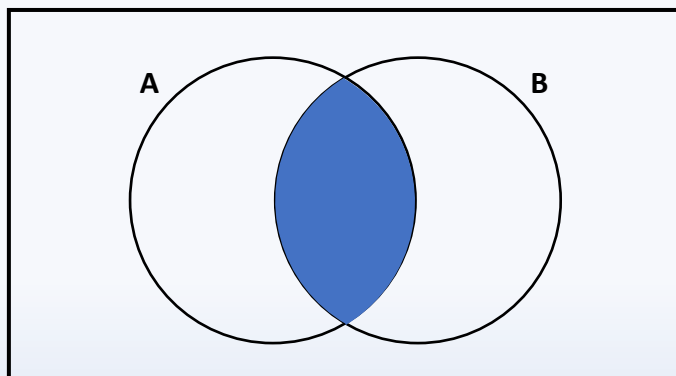
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Set Theory: Operations, Intersection



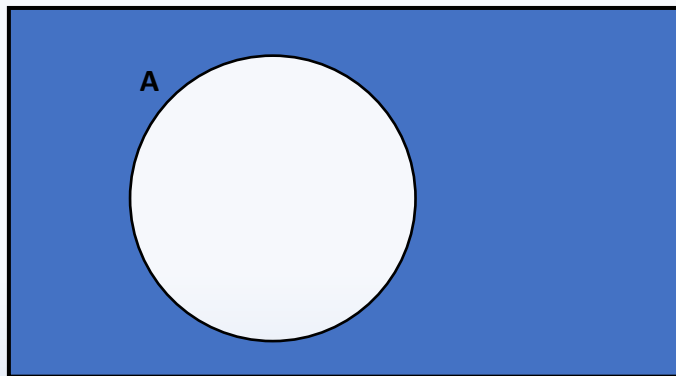
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Set Theory: Operations

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Set Theory: Operations, Complement



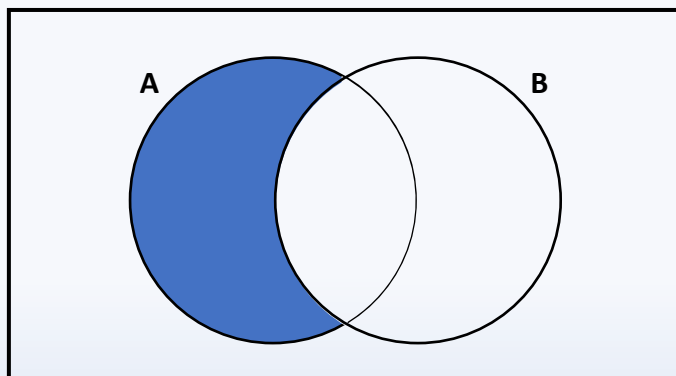
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Set Theory: Operations

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Set Theory: Operations, Relative Complement



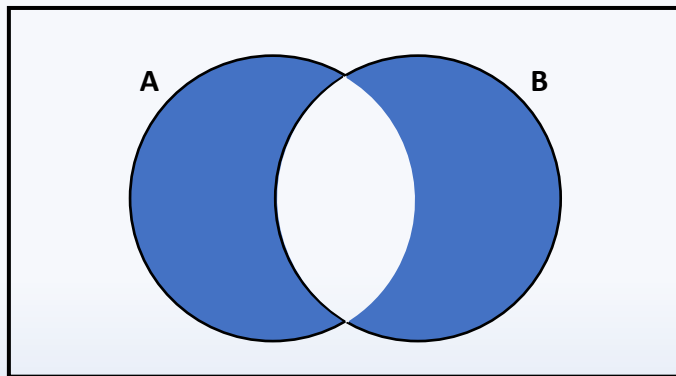
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Set Theory: Operations

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Set Theory: Operations, Symmetric Difference



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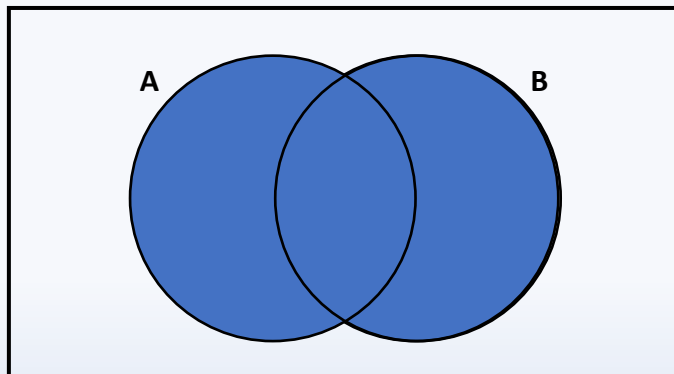
Set Theory: Operations



- What does $(A \cup B) - (A \cap B)$ equal?
 - A. Union: $A \cup B = \{x: x \in A \vee x \in B\}$
 - B. Intersection: $A \cap B = \{x: x \in A \wedge x \in B\}$
 - C. Complement: $A' = \{x: x \notin A\}$
 - D. Reflexive Complement: $A - B = \{x: x \in A \wedge x \notin B\}$
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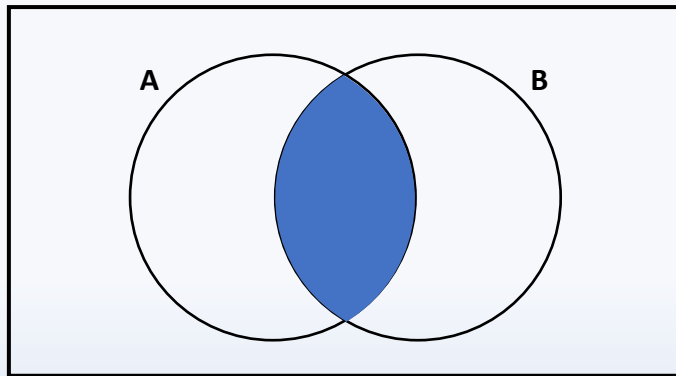
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Set Theory: Operations, Union



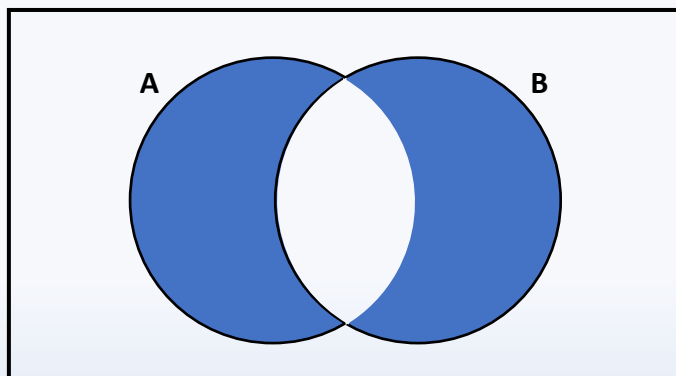
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Set Theory: Operations, Intersection



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Set Theory: Operations, Symmetric Difference



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Set Theory: Relations

- $A \subseteq B$: A is a **subset** of B, if and only if (iff) $a \in A \Rightarrow a \in B$.
- $A \subset B$: A is a **proper subset** of B, iff $A \subseteq B \wedge B - A \neq \emptyset$.
- $A = B$: A is **equal** to B, iff $A \subseteq B \wedge B \subseteq A$
- **Partitions** of A are sets $A_1, A_2, A_{\dots}, A_n$ such that:
 - $A_1 \cup A_2 \cup A_{\dots} \cup A_n = A$, and
 - $A_i \cap A_j = \emptyset$, for any i, j

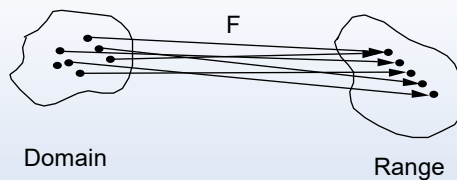
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Functions

- **Domain**: Input to the Function (e.g., A)
- **Range**: Output from the function (e.g., B)

$$f: A \rightarrow B$$

$$f \subseteq A \times B$$



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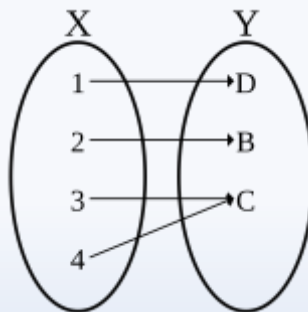
Functions: Types

- f is a *function* from X **onto** Y iff $f(X) = Y$
- f is a *function* from X **into** Y iff $f(X) \subset Y$
- f is a **one-to-one** *function* from X to Y iff,
for all $a, b \in X, a \neq b \Rightarrow f(a) \neq f(b)$
- f is a **many-to-one** *function* from X to Y iff,
there exists $a, b \in X, a \neq b \Rightarrow f(a) = f(b)$

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Functions: Types

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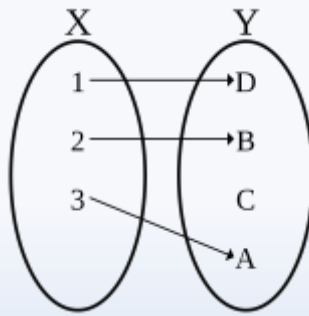


Every element of Y is covered by $f(X)$

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Functions: Types

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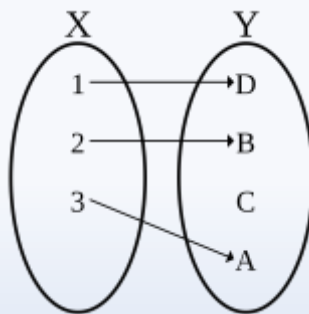


At least one element of Y is not covered by $f(X)$

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Functions: Types

- f is a **one-to-one** function from X to Y iff,
for all $a, b \in X, a \neq b \Rightarrow f(a) \neq f(b)$

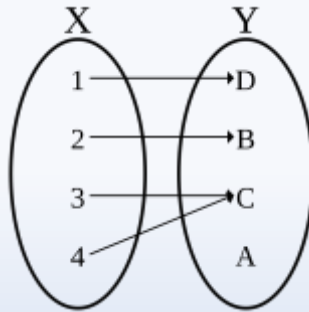


Every element in X maps (via f) to a unique Y

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Functions: Types

- f is a **many-to-one** function from X to Y iff,
there exists $a, b \in X, a \neq b \Rightarrow f(a) = f(b)$



Not every element in X maps (via f) to a unique Y

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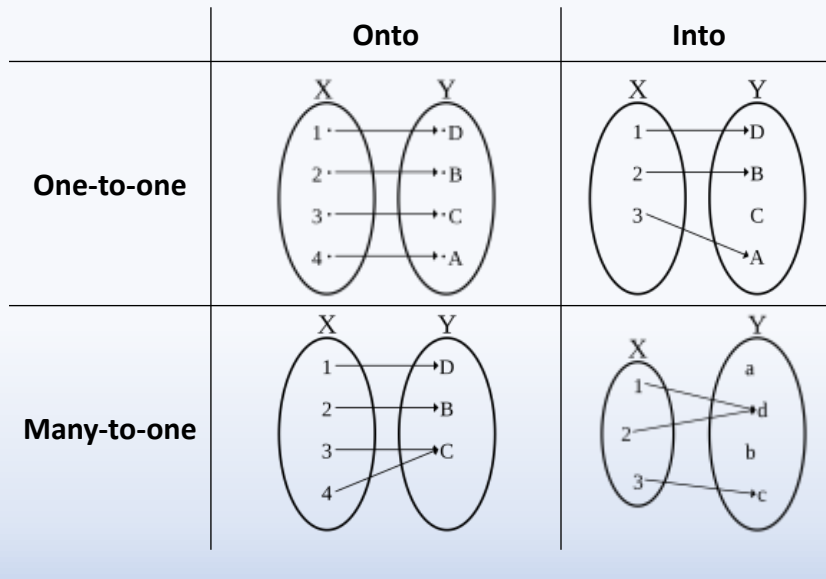
In Class Exercise

Show an example of each of the following possibilities.

	Onto	Into
One-to-one		
Many-to-one		

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In Class Exercise



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